TASK 2 | OFFICE 1 INSTALLATION DETAILS AND DESIGN CONDITIONS

A3 LANDSCAPE | NOT TO SCALE | DIMENSIONS IN mm | OFFICE 1

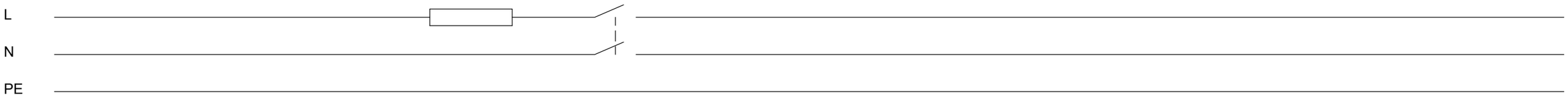
1. EVERY POWER ASSEMBLY HAS AN UPSTREAM FCU: SOCKETS13 A / WH01 10 A

Accessible main/ring junction

DP switched FCU at tap
13 A sockets /10 A WH01

4 mm2 L/N/CPC spur
0.5m assumed

ONE assembly: FB / SO / WH01
Sockets share13 A; WH01 load6.364 A



For rings: both L/N/CPC ends return to their own DB terminals. FCU is before the spur; no unfused extra taps. Main terminals must accept 6/10 mm2.

FB01-09:13 A FCU per four-socket floorbox. SO01-03:13 A FCU per twin socket. FS01:10 A DP FCU for WH01.

2. SEPARATE CONTAINMENT BANKS - OVERLAPPING PLAN LINES ARE ROUTING CORRIDORS

Lighting LC-A..G	A: P1/P2	B: P3/P4	C: P5/P6/P8	P7 alone	Feeder alone
7 circuits / Cg0.54	4 ring legs / Cg0.65	4 ring legs / Cg0.65	3 radials / Cg0.70	Cg1.00	Cg1.00

Method B, no insulation contact. Keep L/N/CPC of each circuit together. Between banks provide horizontal cable clearance >2x largest adjacent cable OD

(EMSD A5(3) Note2); separate enclosures and verify cable clearance. A partition alone is not thermal separation. If grouping changes, recalculate sizes.

3. A1 RING DESIGN BASIS AND EQUIPMENT SPECIFICATION

EMSD 6C/6E, Table6(1): B32; 6 mm2 copper; corrected leg Iz23.1855 A >=20 A; each service envelope <=100m2; intact L/N/CPC rings; two fused taps per ring.

Ring leg currents14.658-15.203 A support load distribution; overload protection follows the A1 standard arrangement. Verify ring continuity before energisation.

TN-S source Ze0.20 ohm/PFC<=6 kA are design conditions. Main and RCBO Icn10 kA. Required total I2t: B10<=70000; B32/FCU<=90000; C63<=1000000 A2s.

Select apparatus meeting the stated breaking capacity and total clearing energy. Control20 AX/LED>=600 W; confirm driver inrush. PE bypasses switching.

4. LIGHTING DESIGN BASIS

92 panels /2256 W. LDT calculation, MF0.80, luminaires1.9m above workplane: mean direct lux open592.9 /pantry546.3 /upper east588.5 /south east662.6.

South east: four38 W panels,4685.55 lm each; mean589.2 lx at2.2m above workplane,17.8% above500 lx target. Lighting positions and routes retained.

Adopted height1.9m and MF0.80 with periodic cleaning. Direct-light calculation; see report for room geometry and perimeter illuminance.